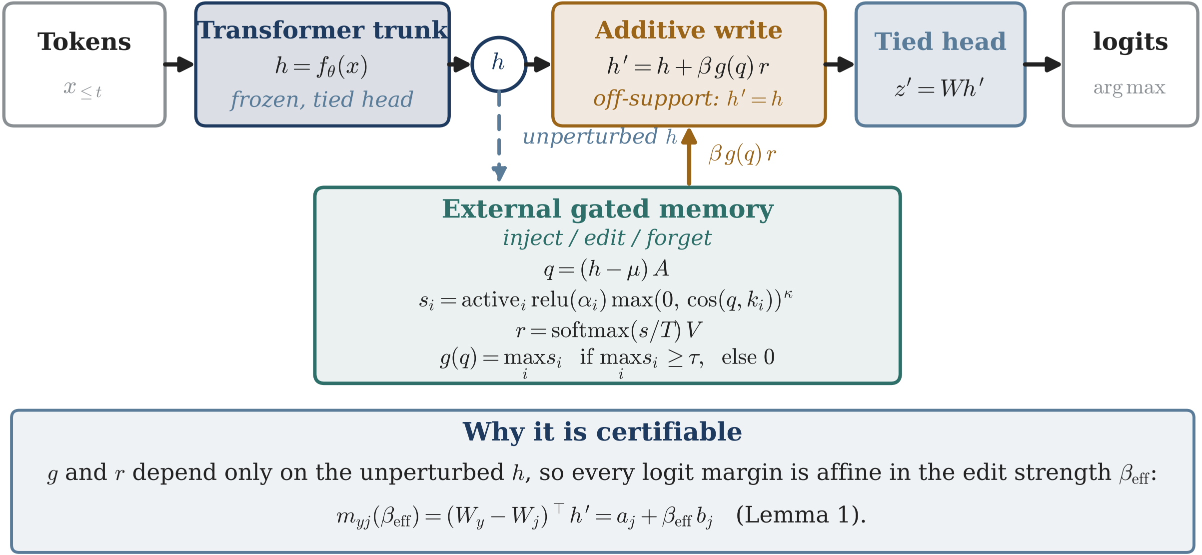
Tokens
$x_{\leq t}$
Transformer trunk
$h = f_\theta(x)$
frozen, tied head
h
unperturbed h
Additive write
$h' = h + \beta\, g(q)\, r$
off-support: $h' = h$
$\beta\, g(q)\, r$
Tied head
$z' = Wh'$
logits
$\arg\max$
External gated memory
inject / edit / forget
$q = (h - \mu)\, A$
$s_i = \mathrm{active}_i\, \mathrm{relu}(\alpha_i)\, \max(0,\, \cos(q, k_i))^\kappa$
$r = \mathrm{softmax}(s/T)\, V$
$g(q) = \max_i s_i \quad \text{if } \max_i s_i \geq \tau, \quad \text{else } 0$
Why it is certifiable
g and r depend only on the unperturbed h, so every logit margin is affine in the edit strength β_{eff}:
$m_{yj}(\beta_{\mathrm{eff}}) = (W_y - W_j)^\top h' = a_j + \beta_{\mathrm{eff}}\, b_j$ (Lemma 1).